

OSTMARK, Sweden

WMO: 024000

Lat: 60.350N

Lon: 12.650E

Elev: 310

StdP: 97.66

Time Zone: 1.00 (EUC)

Period: 86-99

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.9	-17.9	-23.5	0.5	-20.8	-20.4	0.6	-17.6	15.4	3.1	14.7	3.1	0.7	320	0.682

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	25.4	16.3	23.6	15.3	21.8	14.3	17.4	23.3	16.3	21.7	15.2	20.0	3.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.3	11.3	18.4	14.1	10.4	17.9	12.9	9.6	16.6	50.0	23.1	46.6	21.9	43.5	20.0	22.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.8	10.2	7.5	DB	-22.6	27.2	5.5	2.3	-26.5	28.8	-29.8	30.1	-32.9	31.4	-36.9	33.0	(4)
(5)			WB	-22.8	18.1	5.4	2.1	-26.7	19.6	-29.9	20.9	-33.0	22.1	-36.9	23.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec			
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)			
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	3.9	-5.0	-4.5	-1.1	2.5	8.2	12.3	15.3	12.9	8.7	3.6	-1.3	-4.9	(6)	
(7)		DBStd	8.25	6.27	5.75	4.18	3.60	3.75	3.20	2.90	2.90	3.07	3.71	4.25	5.24	(7)	
(8)		HDD10.0	2600	466	407	346	226	81	11	1	6	60	198	339	461	(8)	
(9)		HDD18.3	5264	724	640	604	474	313	182	103	169	290	456	589	719	(9)	
(10)		CDD10.0	389	0	0	0	2	26	81	163	97	20	1	0	0	(10)	
(11)		CDD18.3	11	0	0	0	0	0	2	8	2	0	0	0	0	(11)	
(12)		CDH23.3	175	0	0	0	0	4	48	104	19	0	0	0	0	(12)	
(13)		CDH26.7	17	0	0	0	0	0	7	10	0	0	0	0	0	(13)	
(14)	Wind	WSAvg	3.0	3.3	3.3	3.4	3.0	3.2	3.1	3.0	2.7	2.9	2.9	2.6	2.9	(14)	
(15)	Precipitation	PrecAvg	836	59	42	39	48	64	89	95	98	75	84	78	65	(15)	
(16)		PrecMax	1183	139	81	89	114	183	147	211	165	165	221	226	112	(16)	
(17)		PrecMin	684	13	13	5	9	5	7	35	31	21	22	31	10	(17)	
(18)		PrecStd	111	29	19	20	28	39	34	37	38	32	41	44	29	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.8	7.5	10.5	18.0	24.0	27.3	28.6	25.3	20.8	13.2	9.4	5.5	(19)	
(20)			MCWB	3.4	4.8	5.6	10.5	13.8	17.5	17.7	15.7	14.6	10.2	7.5	3.9	(20)	
(21)		2%	DB	4.4	5.6	7.4	13.9	21.9	24.4	26.2	23.6	17.4	11.2	7.7	4.2	(21)	
(22)			MCWB	2.5	3.6	3.3	7.8	12.9	15.7	16.9	15.5	12.2	8.6	5.8	3.1	(22)	
(23)		5%	DB	3.2	4.1	5.5	11.1	19.6	22.1	24.3	21.2	15.5	10.1	5.8	3.1	(23)	
(24)			MCWB	1.6	2.6	2.4	6.1	11.7	13.9	16.1	14.4	11.2	8.0	4.7	2.1	(24)	
(25)		10%	DB	2.0	2.7	4.0	8.9	16.6	19.8	22.2	18.8	13.9	9.0	4.6	1.8	(25)	
(26)			MCWB	0.8	1.3	1.7	4.9	10.0	12.8	15.0	13.6	10.5	7.2	3.7	0.9	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.1	5.2	6.5	11.0	14.7	18.0	19.0	17.7	15.7	11.1	8.0	4.5	(27)	
(28)			MCDWB	5.1	6.5	9.3	17.1	22.0	25.8	26.0	21.7	18.6	12.4	9.2	5.1	(28)	
(29)		2%	WB	3.0	4.0	4.2	8.5	13.4	16.5	17.9	16.5	13.7	9.6	6.1	3.4	(29)	
(30)			MCDWB	4.0	5.1	5.9	12.3	20.9	23.2	24.0	20.6	15.9	10.5	7.2	4.1	(30)	
(31)		5%	WB	2.0	2.8	3.1	6.9	12.0	14.7	16.9	15.5	12.4	8.6	4.9	2.3	(31)	
(32)			MCDWB	2.8	4.0	4.5	10.6	18.9	20.5	23.0	19.8	14.5	9.5	5.7	2.9	(32)	
(33)		10%	WB	0.9	1.4	2.1	5.4	10.5	13.4	15.7	14.3	11.0	7.6	3.8	1.0	(33)	
(34)			MCDWB	1.7	2.4	3.9	8.0	16.3	18.6	20.9	18.0	13.1	8.7	4.6	1.6	(34)	
(35)	Mean Daily Temperature Range			MDBR	5.2	6.5	7.2	8.8	11.3	10.1	10.9	9.2	8.7	5.7	4.5	5.3	(35)
(36)		5% DB	MCDBR	5.5	5.3	8.3	12.9	16.3	15.4	14.1	12.9	11.3	6.8	5.5	5.6	(36)	
(37)			MCWBR	5.2	4.5	5.5	7.4	8.2	7.4	6.4	5.3	6.2	5.0	4.9	5.4	(37)	
(38)		5% WB	MCDWB	5.1	5.2	6.9	12.2	15.8	14.0	12.4	10.2	8.1	5.2	5.2	5.4	(38)	
(39)			MCWBR	5.1	4.9	5.0	7.0	8.4	7.2	6.3	5.2	4.8	4.6	4.9	5.3	(39)	
(40)	Clear-Sky Solar Irradiance	taub		0.242	0.286	0.312	0.350	0.351	0.342	0.362	0.353	0.323	0.310	0.262	0.215	(40)	
(41)		taud		2.208	2.306	2.356	2.378	2.439	2.484	2.441	2.462	2.536	2.488	2.291	2.103	(41)	
(42)		Ebn at Noon		544	721	821	849	877	889	859	839	811	695	514	411	(42)	
(43)		Edh at Noon		38	64	84	99	100	98	100	90	71	54	35	26	(43)	
(44)	All-Sky Solar Radiation	RadAvg		0.28	0.93	2.42	3.82	4.75	5.42	4.83	3.79	2.52	1.13	0.36	0.17	(44)	
(45)		RadStd		0.04	0.18	0.27	0.48	0.47	0.34	0.67	0.40	0.31	0.22	0.10	0.02	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (56 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page